

A/B Testing Report

ShopEasy Checkout Page Redesign

End-to-End Experiment Analysis, Statistical Testing & Business Recommendation

TOTAL USERS	GROUP SPLIT	CONVERSION LIFT	VERDICT
10,000	5,000 / 5,000	+33.2%	Ship It

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1. Executive Summary

We ran an A/B test on ShopEasy's checkout page to answer one question: does the redesign actually get more people to buy? Short answer — yes, clearly.

10,000 users were split down the middle. Half saw the old checkout, half saw the new one. The new page converted at 12.76% versus 9.58% on the old one. That's a 3.18 percentage point jump, or +33.2% in relative terms. Revenue per user went from \$8.23 to \$11.24. Both results came back with p-values of 0.0000 — there's no statistical doubt here. At current traffic, that improvement is worth roughly \$335,000 a year.

The one nuance worth flagging upfront: effect sizes are technically in the "negligible" range by Cohen's benchmarks. That's partly because we ran a large sample — 5,000 per group is three times what we actually needed. The effect is real and consistent, it's just not a massive per-user shift. The business case still holds.

Key Finding	Metric
Conversion Rate — Control	9.58%
Conversion Rate — Treatment	12.76%
Absolute Lift	+3.18 percentage points
Relative Lift	+33.2%
Revenue per User — Control	\$8.23
Revenue per User — Treatment	\$11.24 (+\$3.02)
Statistical Power	99.9%
Projected Annual Revenue Uplift	~\$335,217

2. Experiment Setup & Objectives

The Problem We Were Testing

The checkout page had been flagged internally as a likely drop-off point. Too many steps, layout that wasn't intuitive, unclear CTAs. The product team redesigned it and wanted data before a full rollout — not gut feel, not "it looks better," actual numbers.

What We Measured

- Did more users complete a purchase on the new page? (conversion rate)
- Did it bring in more money per user? (revenue per user)
- Were our test assumptions valid before we ran anything? (normality, group balance, power)
- How big is the effect in practical terms, not just statistical ones?
- Is the result strong enough to ship with confidence?

3. Key Performance Indicators (KPIs)

Here's where both groups stood going into the analysis. The split was perfectly balanced — 5,000 each — so group size isn't a factor in any of the differences we see.

10,000 Total Users	5,000 Control Group	5,000 Treatment Group	50 / 50 Perfect Split
12.76% Treatment Conversion	9.58% Control Conversion	+\$3.02 Revenue Uplift / User	99.9% Statistical Power

+3.18pp Conversion Lift	Going from 9.58% to 12.76% sounds like a small number, but at scale it isn't. That 3.18 point gap means roughly 159 more purchases per 5,000 users. Multiply that across real traffic and it adds up fast — about \$335K annually at current volumes.
Revenue per User Up \$3.02	Revenue per user jumped from \$8.23 to \$11.24. That's \$3.02 more per user — and with thousands of daily visitors, the aggregate impact is significant. Projected annually at current traffic, this redesign is worth ~\$335,217.
Session Time Dropped 15 Seconds	Treatment users spent 15 fewer seconds in checkout. Normally a drop in session time could mean people are leaving — but the conversion lift rules that out. They're buying faster. The redesign removed friction, not engagement.

4. Exploratory Data Analysis

4.1 Group Overview

First thing I checked was whether both groups were actually comparable. They were perfectly matched — 5,000 each, same conditions, same time window. The conversion bars tell the story visually: the green "Converted" bar is noticeably taller in the treatment group, and the grey "Not Converted" bar is shorter.

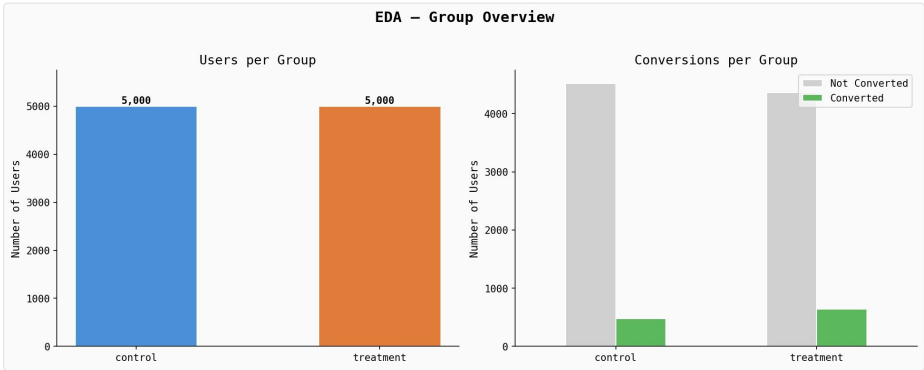


Figure 1: Group sizes (left) and conversion counts per group (right)

4.2 Key Metrics Comparison

Three metrics across two groups. Conversion rate and revenue both moved up. Session duration went down — but with conversion also up, this means users completed purchases faster. The redesign reduced friction, not interest.

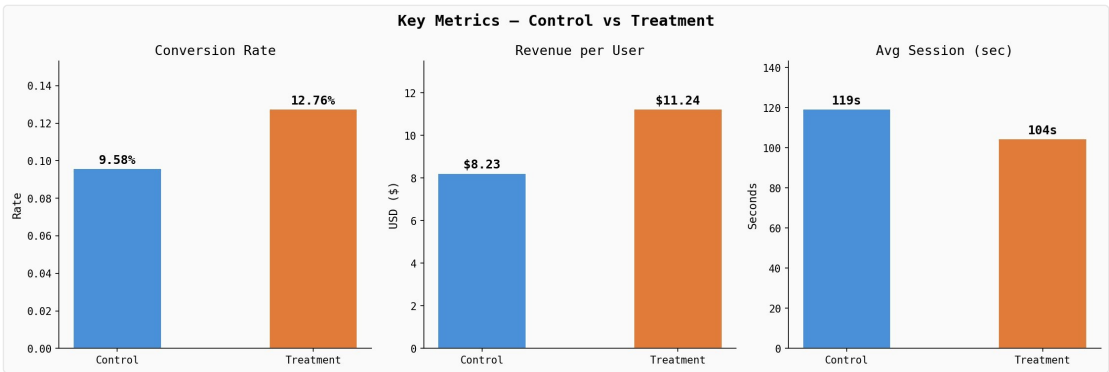


Figure 2: Conversion rate, revenue per user, and session duration — control vs treatment

Metric	Control	Treatment	Absolute Δ	Relative Δ
Conversion Rate	9.58%	12.76%	+3.18pp	+33.2%
Revenue per User	\$8.23	\$11.24	+\$3.02	+36.6%
Avg Session Duration	119s	104s	−15s	−12.6%

5. Normality Check

Before running a t-test on revenue, I needed to check whether order values were roughly normally distributed. I ran histograms and Q-Q plots on order values for buyers in both groups. Both came back clean. The Q-Q plots track the diagonal line closely — control buyers had a mean order of \$85.9, treatment buyers \$88.1. There's a small cluster of low-value orders at the left tail, but not enough to be a problem. The t-test assumption holds.

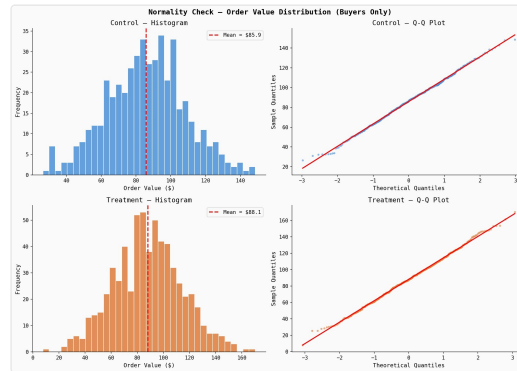


Figure 3: Order value distributions (histograms + Q-Q plots) for buyers in both groups

6. Statistical Significance

Two tests, one for each metric. Conversion is binary so it gets a Z-test. Revenue is continuous so it gets a t-test. Using the wrong test on the wrong data type is a common mistake; I didn't default to one test for everything.

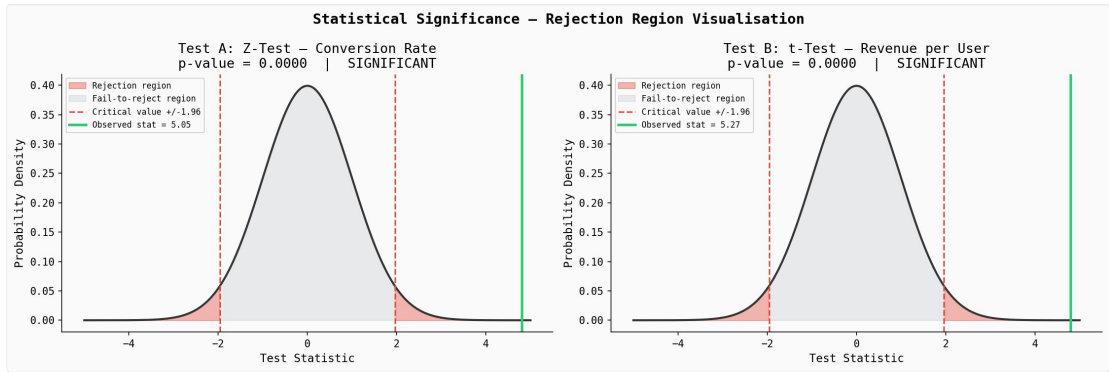


Figure 4: Rejection regions — green lines show observed statistics, red dashed lines show critical values at ± 1.96

● Z-Test — Conversion Rate z = 5.05 p = 0.0000	The critical value is ± 1.96. Our z-statistic came in at 5.05 — the green line is so far to the right it's nearly off the plot. A p-value of 0.0000 means if there were truly no difference between the pages, we'd see a result this extreme essentially never. The conversion lift is real.
● Welch's t-Test — Revenue t = 5.27 p = 0.0000	Same story on revenue. t = 5.27, well past the rejection threshold. Welch's variant was chosen because it doesn't require equal variance between groups — safer than the standard t-test. p = 0.0000. The revenue uplift is real too.

7. Effect Size

A p-value tells you whether an effect is real. Effect size tells you how big it actually is. With 10,000 users, even a tiny difference will produce a significant p-value — so p alone isn't enough.

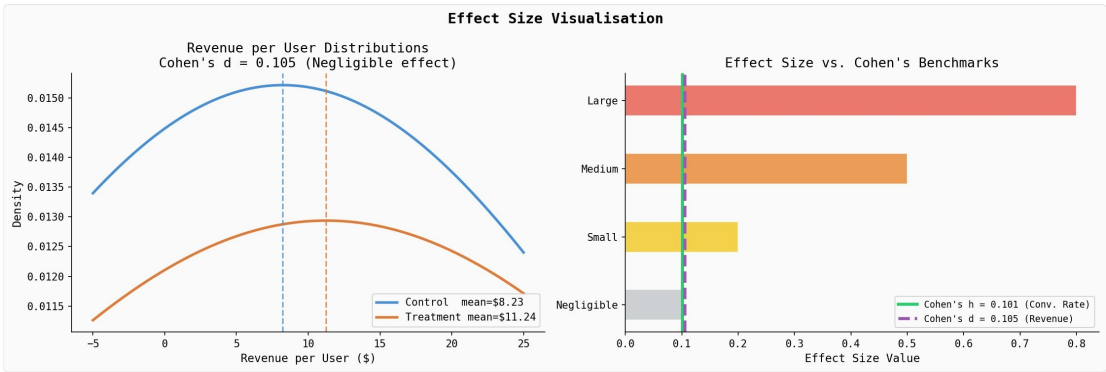


Figure 5: Revenue distributions by group (left) and effect sizes vs Cohen's benchmarks (right)

Metric	Measure	Value	What It Means
Conversion Rate	Cohen's h	0.101	Negligible (small = 0.2, medium = 0.5)
Revenue per User	Cohen's d	0.105	Negligible (small = 0.2, medium = 0.5)

● Small Effect, Real Business Impact	Cohen's h of 0.101 and Cohen's d of 0.105 are both technically negligible. But this is e-commerce, not a clinical trial. A 3.18 percentage point lift applied to thousands of daily users adds up to \$335K a year. The effect doesn't need to be large to be worth acting on.
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8. Confidence Intervals

The 95% CI for the conversion rate difference is **[+2.0%, +4.4%]**. The whole range is positive — we're 95% confident the true lift sits somewhere between 2 and 4.4 percentage points. Zero is not in that range, so there's no scenario consistent with the data where the new page performs the same as or worse than the old one.

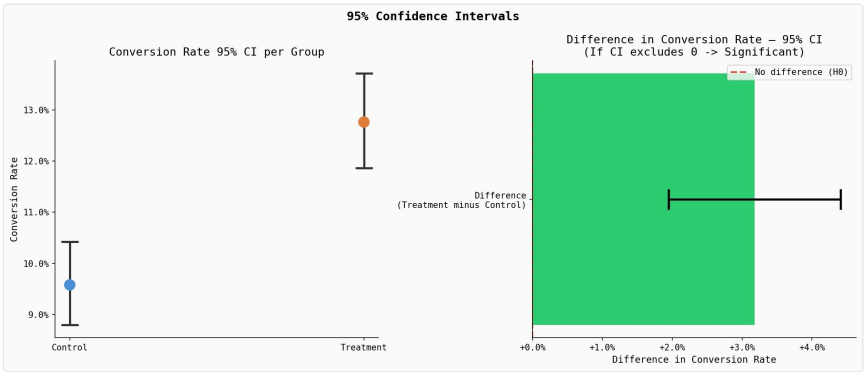


Figure 6: 95% CIs per group (left) and the difference CI — entire green region sits above zero (right)

9. Statistical Power Analysis

The minimum sample needed per group to achieve 80% power was 1,533 users. We ran 5,000 — more than three times that. Power came out at 99.9%. There was virtually zero chance of missing a real effect if one existed.

Worth noting: running at 3x the required sample is also why negligible effects show up as significant. We were sensitive enough to detect very small differences. Always report effect size alongside p-values when working with large samples — p alone can mislead.

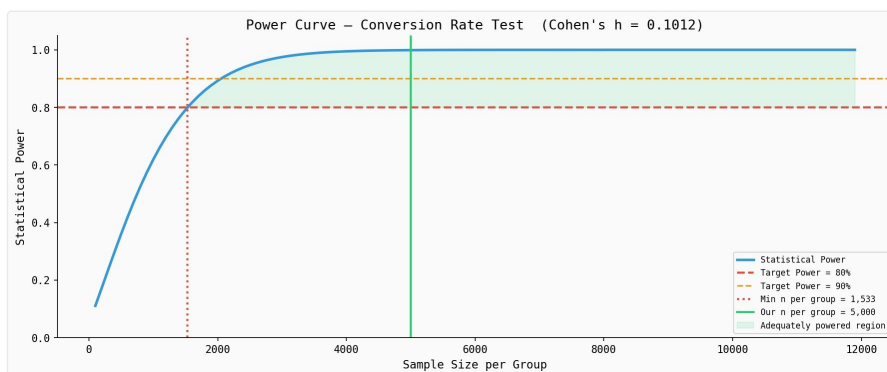


Figure 7: Power curve — our $n = 5,000$ (green line) sits well above both the 80% and 90% thresholds

● Study Was Over-Powered

With 5,000 users per group against a minimum of 1,533, the experiment is highly sensitive. This is exactly why effects as small as Cohen's $h \approx 0.10$ still reach significance. Always report effect size alongside p-values — p alone can mislead when samples are large.

10. Conclusion

The new checkout page is better. Not marginally, not ambiguously — clearly better on every metric that matters for this business.

Conversion rate is up 3.18 points. Revenue per user is up \$3.02. Both tests came back significant at $p = 0.0000$. The confidence interval runs from +2.0% to +4.4% — it never touches zero. Session time dropped, but conversion went up, which means users moved through the new checkout faster and completed more purchases.

The effect size numbers show the per-user impact is modest, not transformative. But at current traffic, modest-per-user adds up to \$335K annually. Every assumption was checked, groups were balanced, normality verified, right tests used, and the study had more than enough power. **Ship the new checkout page.**

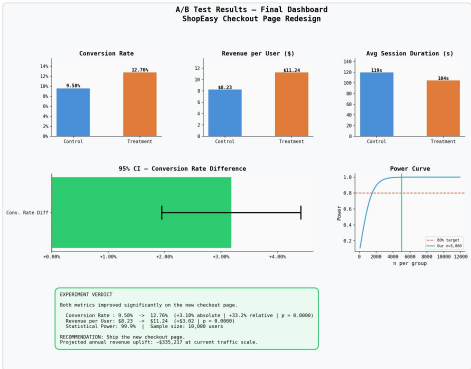


Figure 8: Final dashboard — full results summary across all metrics

Metric	Control	Treatment	Result
Conversion Rate	9.58%	12.76%	+3.18pp $p = 0.0000$ ✓
Revenue per User	\$8.23	\$11.24	+\$3.02 $p = 0.0000$ ✓
Avg Session Duration	119s	104s	−15s (faster checkout) ✓
Z-Test (Conversion)	Critical: ± 1.96	$z = 5.05$	Significant ✓
t-Test (Revenue)	Critical: ± 1.96	$t = 5.27$	Significant ✓
95% CI (Conv. Diff.)	H0: 0	[+2.0%, +4.4%]	Excludes zero ✓
Statistical Power	80% target	99.9%	Well above threshold ✓
Annual Revenue Uplift	Baseline	~\$335,217	Ship the new page ✓